

Voice Scan

- Voice-scan technology utilizes the distinctive aspects of the voice to verify the identity of individuals.
- Voice-scan is occasionally confused with speech recognition:
- It is a technology that translates what a user is saying (a process unrelated to authentication).
- Voice-scan technology, by contrast, verifies the identity of the individual who is speaking.

Voice Scan Vs Speech Recognition

- Speech recognition is used to translate the **spoken word into an account number**,
- voice-scan verifies the **vocal characteristics** against those associated with this account.
- Voice-scan combines elements of behavioral and physiological biometrics:
- While the **shape of the vocal tract** determines to a large degree how a voice sounds,
- a **user's behavior determines what is spoken and in what fashion**.

Voice Scan- strengths

Voice Scan- Weakness

Voice Scan- Components

- It is potentially more susceptible to replay attacks than other biometrics.
- Its accuracy is challenged by low-quality capture devices, ambient noise, and so on.
- The success of voice-scan as a PC solution requires users to develop new habits.

The large size of the template limits the number of potential applications

Components of Voice scan



Components of Voice Scan

Voice-scan systems are similar to facial-scan systems in that they leverage existing **acquisition hardware** and are based primarily on proprietary **software engines that provide template generation and matching**.

Hardware Components : Certain types of acquisition devices, whether **microphones, land telephones, or mobile phones**.

Software Components : A user's spoken phrase is converted from analog to digital format and transmitted to a local or central PC for template generation and storage (during enrollment) or template generation and matching (during verification).

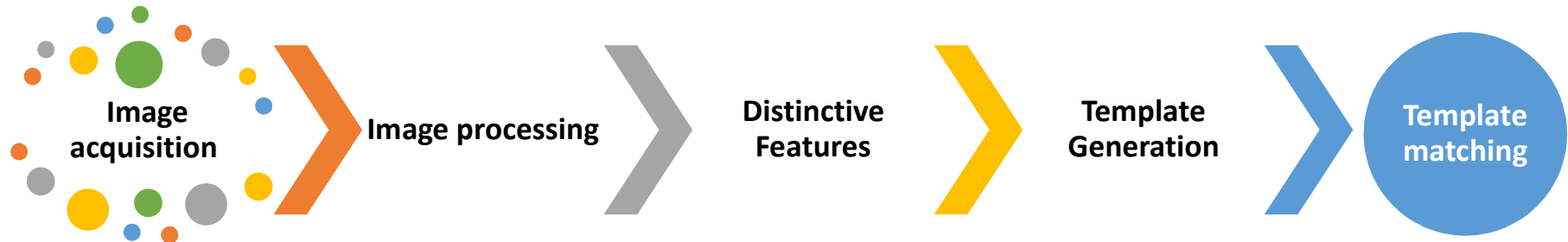
Components of Voice Scan

For desktop verification applications, the engine that provides template-based functions may reside on a local or a central PC, or may be Web-enabled.

In more common telephony applications, the software engine is either located within the institution with which users are interacting (such as a financial institution) or hosted by a third party that offers outsourced biometric verification.



How Voice Scan Technology Works





How Voice Scan Technology Works

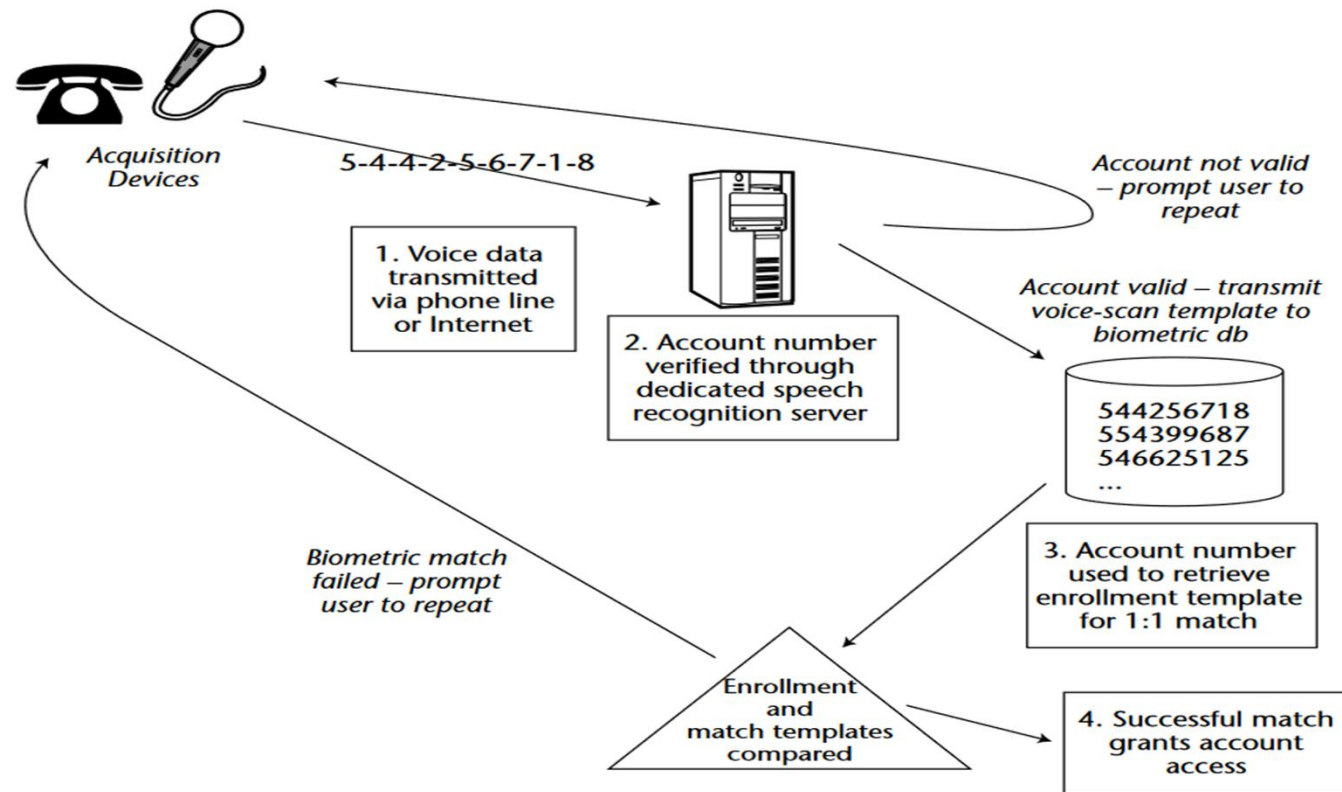
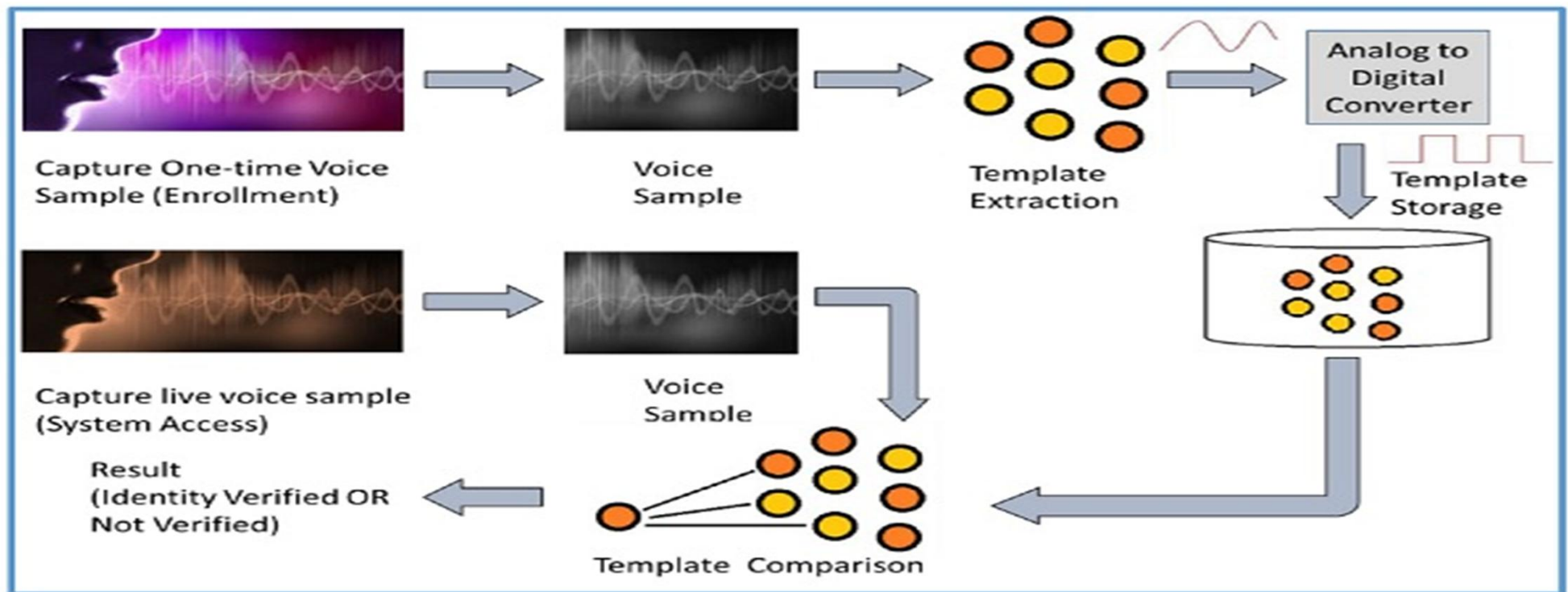


Figure 7.1 Voice-scan and voice recognition—integrated.



How Voice Scan Technology Works



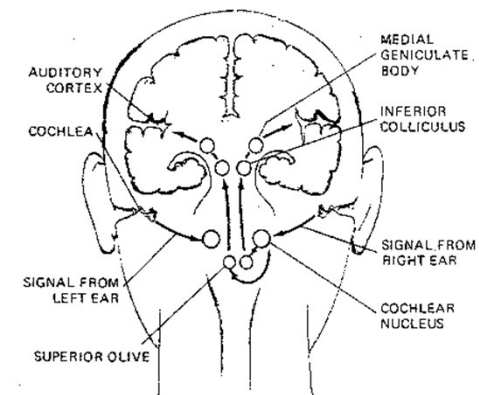
What is the task?

- Getting a computer to understand spoken language
- By “understand” we might mean
 - React appropriately
 - Convert the input speech into another medium, e.g. text

How do humans do it?



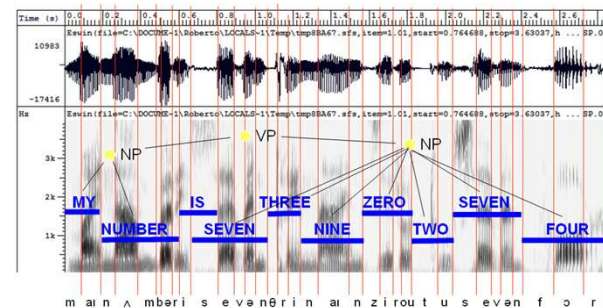
- Articulation produces
- sound waves which
- the ear conveys to the brain
- for processing



How might computers do it?



- Digitization
- Acoustic analysis of the speech signal
- Linguistic interpretation



Speech recognition

What's hard about that?

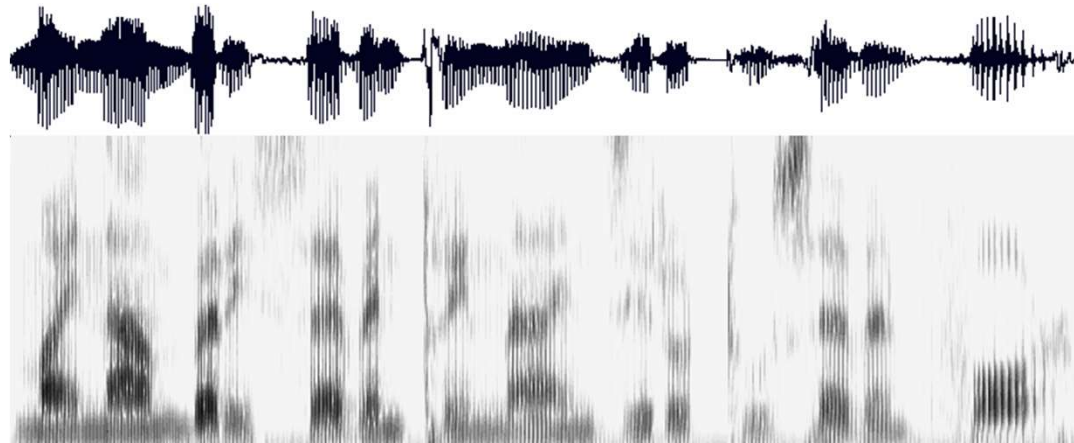
- Digitization
 - Converting analogue signal into digital representation
- Signal processing
 - Separating speech from background noise
- Phonetics
 - Variability in human speech
- Phonology
 - Recognizing individual sound distinctions (similar phonemes)
- Lexicology and syntax
 - Disambiguating homophones
 - Features of continuous speech
- Syntax and pragmatics
 - Interpreting prosodic features
- Pragmatics
 - Filtering of performance errors (disfluencies)

Digitization

- Analogue to digital conversion
- Sampling and quantizing
- Use filters to measure energy levels for various points on the frequency spectrum
- Knowing the relative importance of different frequency bands (for speech) makes this process more efficient
- E.g. high frequency sounds are less informative, so can be sampled using a broader bandwidth (log scale)

Separating speech from background noise

- Noise cancelling microphones
 - Two mics, one facing speaker, the other facing away
 - Ambient noise is roughly same for both mics
- Knowing which bits of the signal relate to speech
 - Spectrograph analysis



Variability in individuals' speech

- Variation among speakers due to
 - Vocal range (f0, and pitch range)
 - Voice quality (growl, whisper, physiological elements such as nasality, adenoidality, etc)
 - ACCENT !!! (especially vowel systems, but also consonants, allophones, etc.)
- Variation within speakers due to
 - Health, emotional state
 - Ambient conditions
- Speech style: formal read vs spontaneous

Speaker-(in)dependent systems

- Speaker-dependent systems
 - Require “training” to “teach” the system your individual idiosyncracies
 - The more the merrier, but typically nowadays 5 or 10 minutes is enough
 - User asked to pronounce some key words which allow computer to infer details of the user’s accent and voice
 - Fortunately, languages are generally systematic
 - More robust
 - But less convenient
 - And obviously less portable
- Speaker-independent systems
 - Language coverage is reduced to compensate need to be flexible in phoneme identification
 - Clever compromise is to learn on the fly

Identifying phonemes

- Differences between some phonemes are sometimes very small
 - May be reflected in speech signal (eg vowels have more or less distinctive f1 and f2)
 - Often show up in coarticulation effects (transition to next sound)
 - e.g. aspiration of voiceless stops in English
 - Allophonic variation

Disambiguating homophones

- Mostly differences are recognised by humans by context and need to make sense

It's hard to wreck a nice beach

What dime's a neck's drain to stop port?

- Systems can only recognize words that are in their lexicon, so limiting the lexicon is an obvious ploy
- Some ASR systems include a grammar which can help disambiguation



How Voice Scan Technology Works

- ❑ Voice Recognition is also called **Speaker Recognition**.
- ❑ At the time of enrollment, the user needs **to speak a word or phrase into a microphone**. This is necessary to acquire speech sample of a candidate.
- ❑ The electrical signal from the microphone is converted into digital signal by an **Analog to Digital (ADC) converter**.
- ❑ It is recorded into the computer memory as a digitized sample.
- ❑ The computer then compares and attempts to match the input voice of candidate with the stored digitized voice sample and identifies the candidate.



Voice Recognition Modalities

There are two variants of voice recognition –

speaker dependent and **speaker independent**.

Speaker dependent voice recognition : It relies on the knowledge of candidate's particular voice characteristics. This system learns those characteristics through voice training (or enrollment).

Speaker Dependent Voice System

- The system needs to be trained on the users to accustom it to a particular accent and tone before employing to recognize what was said.
- It is a good option if there is only one user going to use the system.

Speaker Independent Voice System

- Speaker independent systems are able to recognize the speech from different users by **restricting the contexts of the speech** such as words and phrases.
- These systems are used for automated telephone interfaces.

Speaker Independent Voice System

- They do **not require training the system** on each individual user.
- They are a **good choice to be used by different individuals** where it is not required to recognize each candidate's speech characteristics.

Difference between Voice and Speech Recognition

Speaker Recognition (Voice Recognition)

The objective of voice recognition is to recognize WHO is speaking.

It is used to identify a person by analyzing its tone, voice pitch, and accent.

Speech Recognition

The speech recognition aims at understanding and comprehending WHAT was spoken.

It is used in hand-free computing, map, or menu navigation.

Data Acquisition

- Voice-scan can utilize any audio capture device, including mobile and land telephones and PC microphones.
- The performance of voice-scan systems can vary according to the quality of the audio signal as well as variation between enrollment and verification devices, so acquisition normally takes place on a device likely to be used for future verification.
- Acquisition is more difficult on PCs than on telephones and is more difficult on mobile phones than on landlines.

Data Processing

- Once a sufficient data set has been acquired for enrollment or verification, voice-scan systems process the vocal recordings prior to template creation
- This processing includes **eliminating gaps** at the beginning and end of the recording, so that only the passphrase is used in template generation, and **filtering out non spoken frequencies** such as those found in telephone lines and microphones

Distinctive Features

- Voice-scan technology measures a variety of vocal qualities, many of which are not detectable by human listeners
- In addition to pitch or fundamental frequency, voice-scan algorithms will often measure other variables such as
 - Gain or intensity,
 - short-time spectrum of speech,
 - formant frequencies,
 - linear prediction coefficients,
 - cepstral coefficients,
 - spectrograms (time-frequency-energy patterns), and
 - nasal coarticulation.

Template Creation

- The overall system of generating a voice-scan template usually relies on a statistics-based pattern matching system, most commonly Hidden Markov models
- Hidden Markov models are generalized profiles that are formed through the comparison of multiple samples to find characteristically repeating patterns.